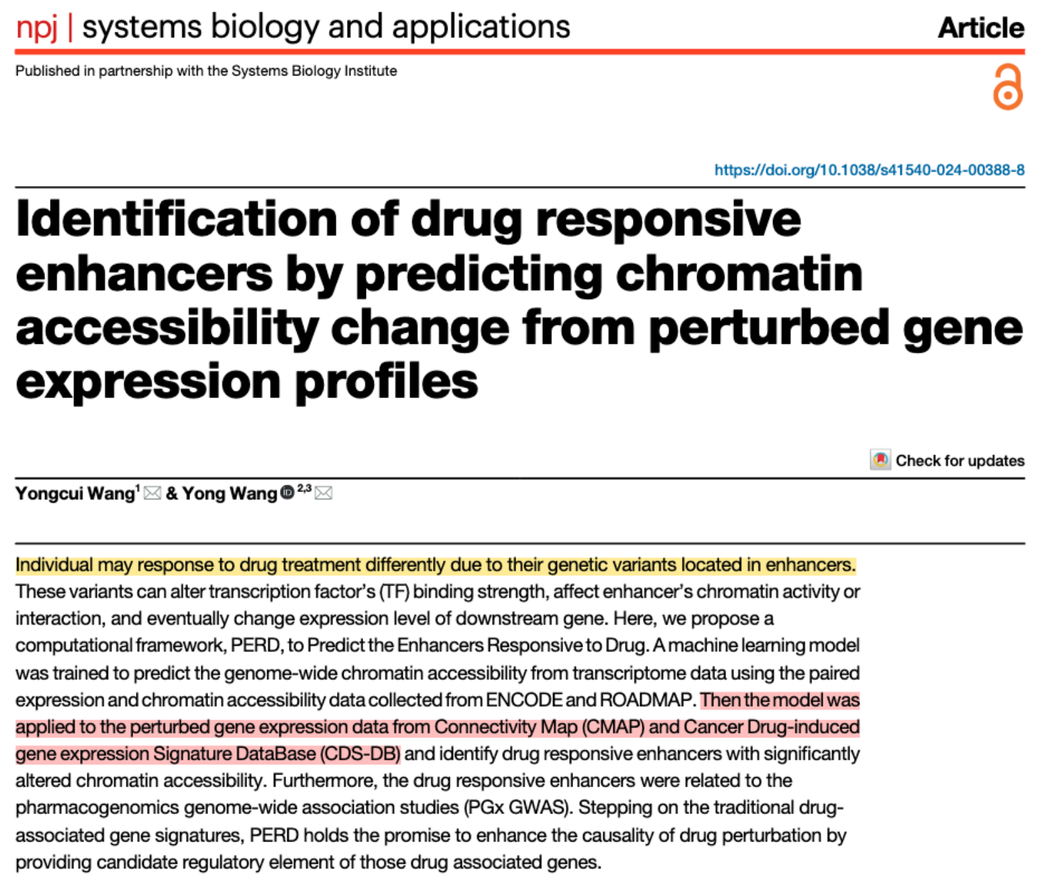
Can RNA predict how a drug changes enhancer accessibility?

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The Journal



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Article

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Identification of drug responsive enhancers by predicting chromatin accessibility change from perturbed gene expression profiles

Check for updates

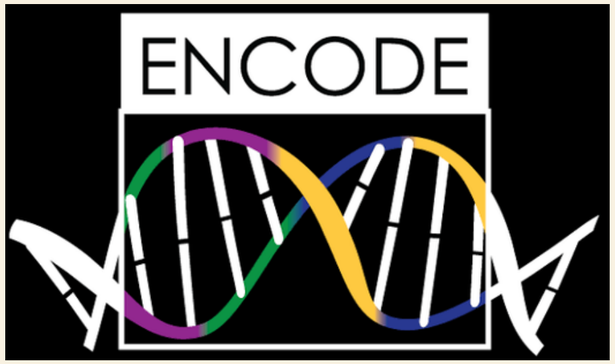
Yongcui Wang[1] & Yong Wang[2,3]

Individual may response to drug treatment differently due to their genetic variants located in enhancers. These variants can alter transcription factor's (TF) binding strength, affect enhancer's chromatin activity or interaction, and eventually change expression level of downstream gene. Here, we propose a computational framework, PERD, to Predict the Enhancers Responsive to Drug. A machine learning model was trained to predict the genome-wide chromatin accessibility from transcriptome data using the paired expression and chromatin accessibility data collected from ENCODE and ROADMAP. Then the model was applied to the perturbed gene expression data from Connectivity Map (CMAP) and Cancer Drug-induced gene expression Signature DataBase (CDS-DB) and identify drug responsive enhancers with significantly altered chromatin accessibility. Furthermore, the drug responsive enhancers were related to the pharmacogenomics genome-wide association studies (PGx GWAS). Stepping on the traditional drug-associated gene signatures, PERD holds the promise to enhance the causality of drug perturbation by providing candidate regulatory element of those drug associated genes.

- Wang, Y. & Wang, Y.
- npj Systems Biology and Applications
- 30 May 2024
- **Build PERD, Predict the Enhancers Responsive to Drug**

Why predict enhancers?

- Most pharmacogenomic GWAS hits sit in regulatory, non-coding DNA (e.g. VKORC1 promoter → warfarin dose).
- To explain them, we need to know which enhancers a drug actually changes.
- But chromatin profiles after drug treatment barely exist, **while drug-perturbed expression profiles are abundant**.



~ 600 cell lines with DNase-seq and RNA-seq paired (not drug-treated)

Connectivity Map (CMAP)

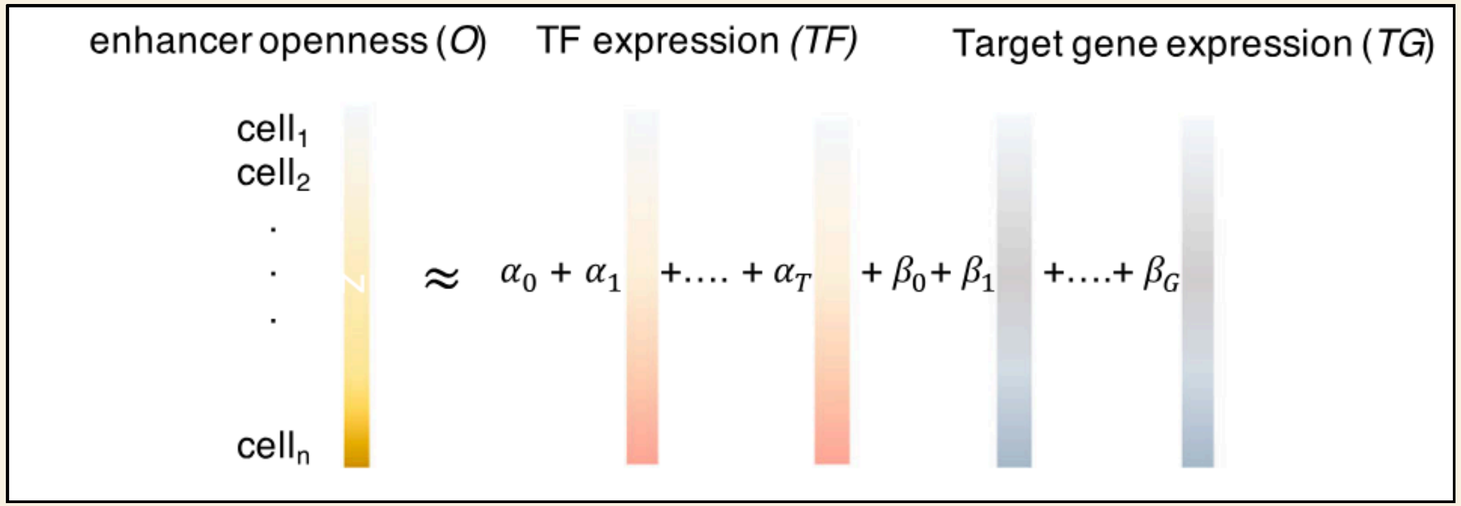
Creating and analyzing large perturbational datasets to aid our understanding of human disease and to accelerate the discovery of novel therapeutics.

CMAP expression profiles: 1,309 compounds, before vs after drug treatment

What PERD does

PERD does two things:

1. **Predicting the enhancer openness** from the related **TF and Target Gene Expression.**
2. Feed the prediction in drug-treated RNA profiles (CMap) and call the enhancers whose predicted openness changes as "**drug-responsive**".

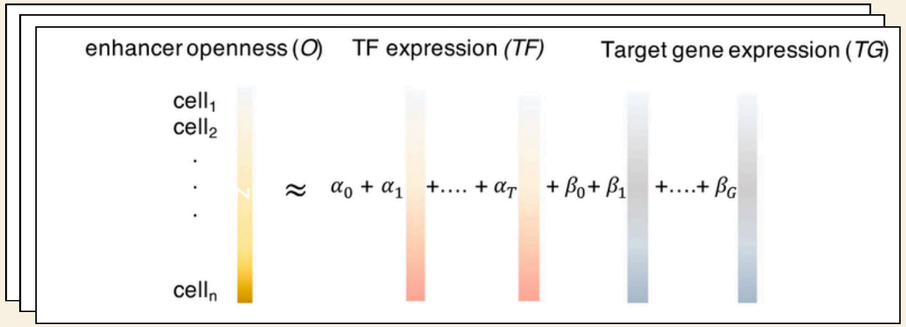


Adapted from Wang & Wang, npj Syst Biol Appl (2024), Fig. 1

How is the model trained?

Samples

54,076 enhancers x 167 paired ENCODE cell lines (110 train, 57 test)

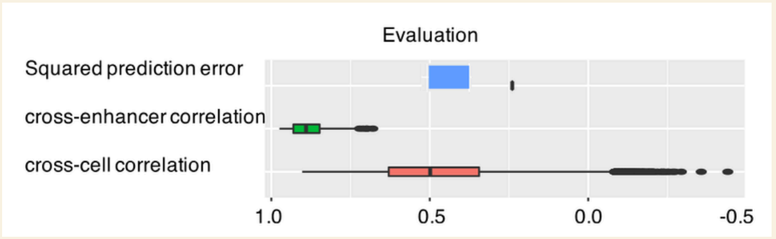


Adapted from Wang & Wang, npj Syst Biol Appl (2024), Fig. 1

Model

Random Forest regression

Evaluation



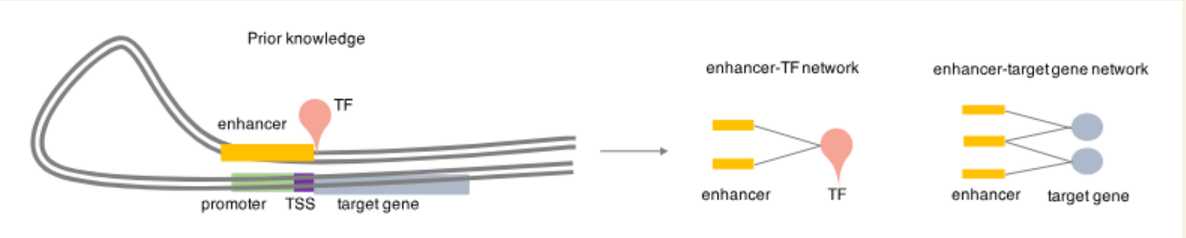
Adapted from Wang & Wang, npj Syst Biol Appl (2024), Fig. 1

Features

Expression of TFs whose ChIP-seq peaks fall in the enhancer, plus its GeneHancer **target genes**

Target

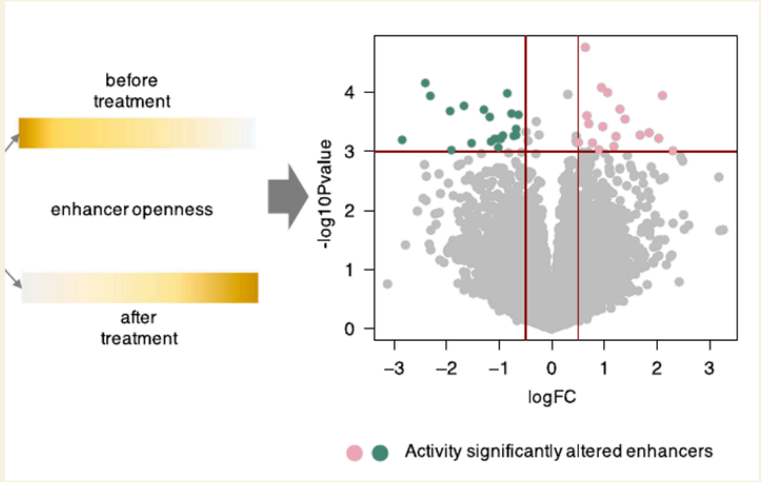
Enhancer openness = max DNase-I signal across the enhancer's genome bins



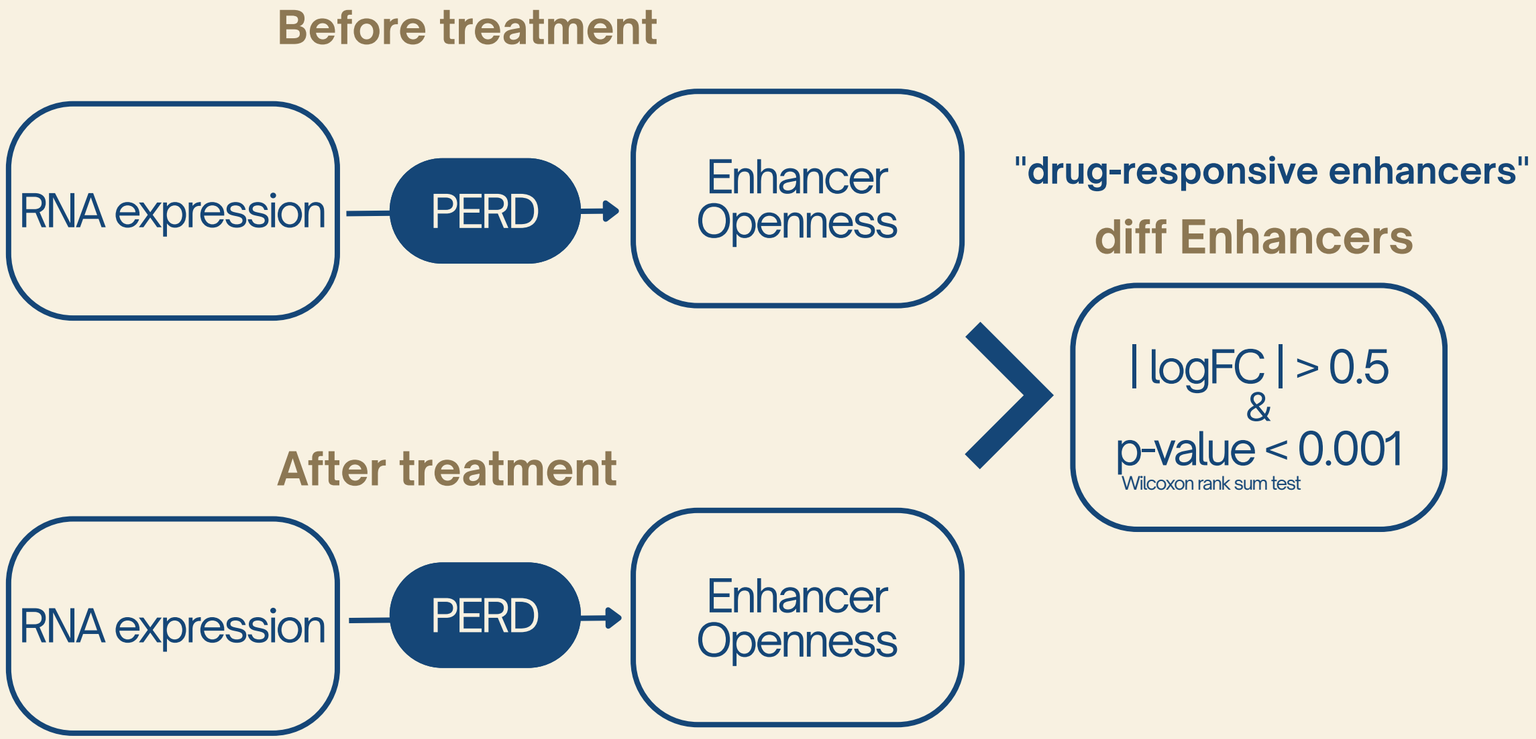
Adapted from Wang & Wang, npj Syst Biol Appl (2024), Fig. 1

How it is used

- Feed the prediction in drug-treated RNA profiles (CMap) and call the enhancers whose predicted openness changes as "**drug-responsive enhancers**".



Adapted from Wang & Wang, npj Syst Biol Appl (2024), Fig. 1



The Gap

PERD calls drug-responsive enhancers, but the calls were never checked against measurements

Improvements

Validation

1. Is PERD a good model of openness in cell types it never saw?
2. Does it predict real drug-induced changes?

Model Improvement

3. Do biologically motivated inputs (TF activity, motifs) help?
4. Does training on real perturbations help?

Is PERD a good model of openness in unseen tissues?

- 167 ENCODE samples → 75 unique tissue/cell types
- **5-fold cross-validation by tissue type**: each sample is predicted by a model that has never seen its tissue
- Stricter than the paper's random 110/57 split, where the same tissue from another donor can be in training
- **Also tested**: one shared model for all enhancers, instead of one model per enhancer

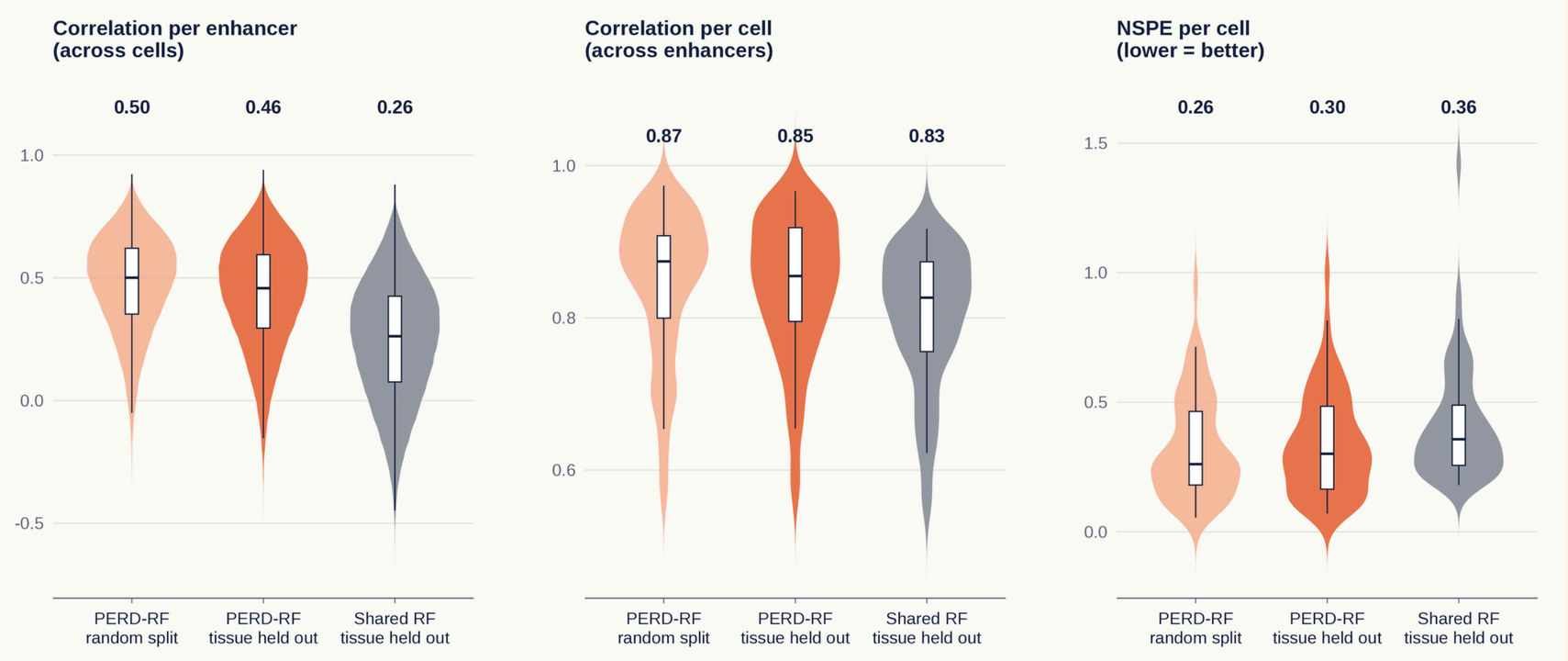
Scored predictions on held-out samples in three metrics:

- **Cross-cell correlation**: for each enhancer, the correlation between predicted and measured openness across held-out samples.
- **Cross-enhancer correlation**: for each sample, the correlation between predicted and measured openness across all 120,100 enhancers.
- **NSPE** (normalized square prediction error).

$$\mathrm{NSPE} = \frac{\sum_e \sum_c (y_{ec} - \hat{y}_{ec})^2}{\sum_e \sum_c (y_{ec} - \bar{y})^2}$$

Is PERD a good model of openness in unseen tissues?

Compare PERD-RF (one model per enhancer) on the paper's random split and with whole tissues held out, against a shared random forest (one model for all enhancers), **all on the same 57 test samples.**



Yes, the result of the tissue held out test are relatively similar with the random split test from the original paper.

Does PERD predict real drug-induced changes?

- Adding five ENCODE perturbation series (16 perturbations, 40 before/after pairs) with RNA and accessibility measured before and after
- Processed exactly like PERD's training data
- PERD-RF predicts openness from the control RNA and from the treated RNA, separately
- NSPE for three things: before (control openness), after (treated openness), and the change (after − before)

Preparing the drug perturbation dataset

16 perturbations, 40 before/after pairs obtained from ENCODE, used in every comparison

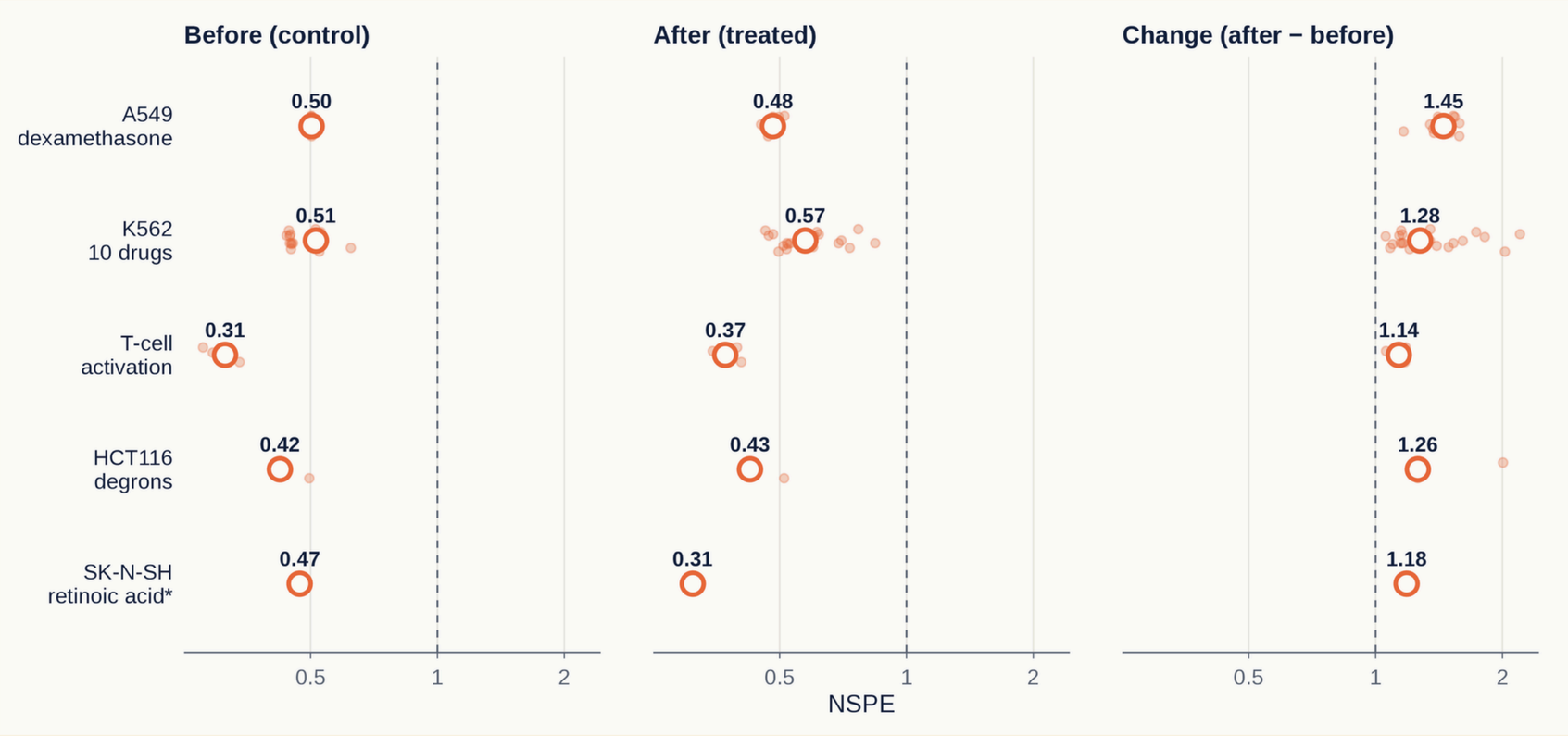
Perturbation	Perturbation count	What it does	Before → after	Pairs used	Assay	In PERD training?
A549 + dexamethasone	1	Steroid drug; switches on the glucocorticoid receptor (GR)	Untreated → 0.5 to 12 h	11 time points	DNase + RNA	No
K562 + 10 drugs	10	Chromatin-enzyme inhibitors (HDAC, BET, p300 …), proteasome and kinase inhibitors	DMSO → drug, 24 / 48 h	20 drug × time	ATAC + RNA	Untreated K562: yes
T cells, activation	1	Anti-CD3/CD28 stimulation, as in an immune response	Resting → activated (same donor)	5 T-cell types	DNase + RNA	Resting T cells: yes
HCT116 degrons	3	Auxin removes one tagged protein within hours: BRD4, CTCF or MED14	Untreated → 6 h auxin	3 proteins	DNase + RNA	No
SK-N-SH + retinoic acid	1	Vitamin A derivative; switches on retinoic-acid receptors (RAR)	Untreated → 48 h	1	DNase + RNA	Treated sample: yes*

16 Pertubations

40 pairs

Applying PERD to the perturbation dataset

PERD knows which enhancers are open, not how much a drug changes them



One dot per pair; circle = median. *SK-N-SH: PERD saw its treated sample.

Before and after: NSPE 0.31–0.57 (well predicted). Change: 1.14–1.45, in all five datasets.

Do biologically motivated inputs help?

Adding **transcription factor activity** and **motifs** to PERD-RF

TF activity
Inferred from each transcription factor's (TF) target genes, not from its own mRNA.

Motif inferred TF activity
Links each enhancer to the TFs whose binding motifs it contains and input its measured TF activity.

Control
Same activity, computed from shuffled target genes, to separate biology from extra inputs.

Validation

- 5-fold predictions for the unseen cell type and using all the 167 ENCODE samples
- 16 Drug-pertubation dataset

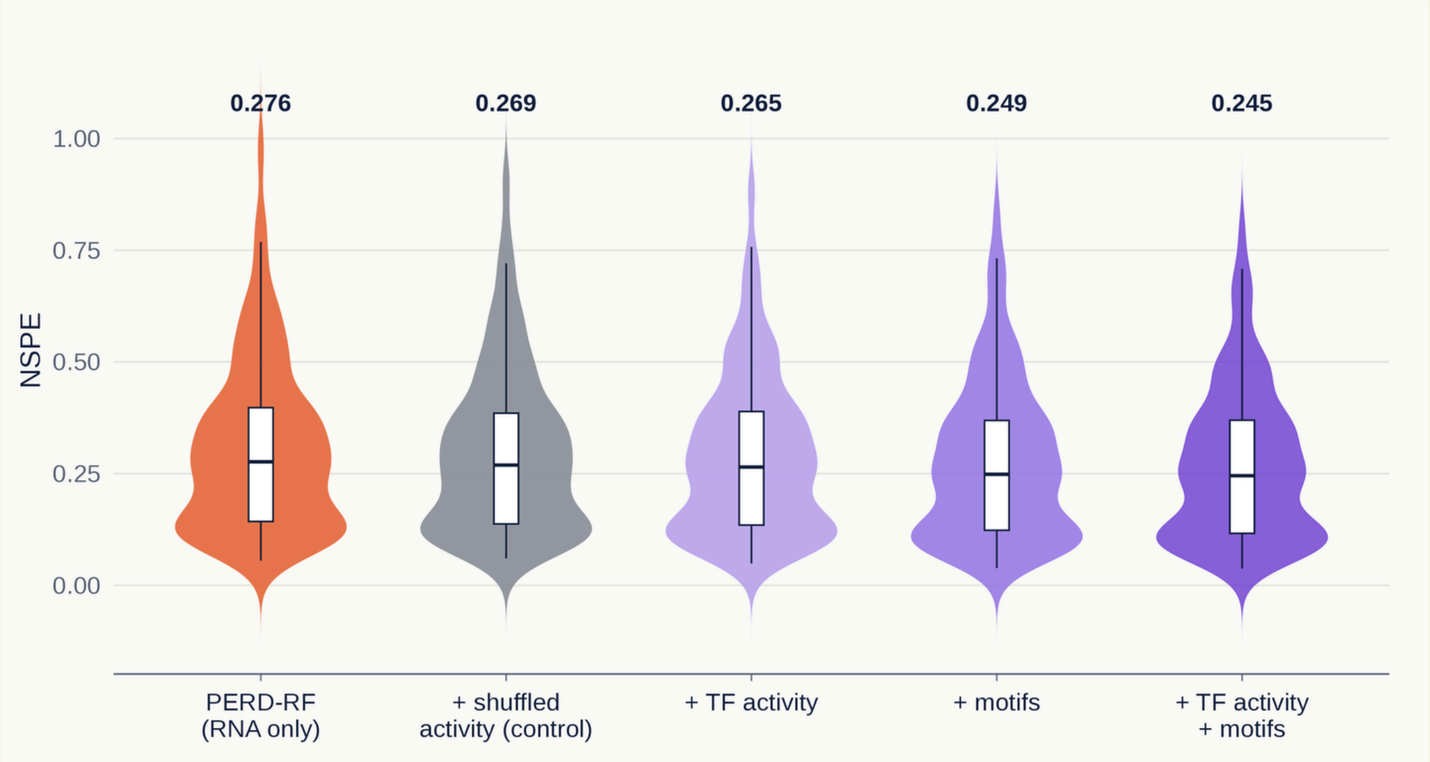
TF–target network: CollecTRI (Müller-Dott et al., Nucleic Acids Res 2023). Activity method: decoupleR ULM (Badia-i-Mompel et al., Bioinformatics Advances 2022). Motif families: Vierstra et al., Nature 2020. Enhancer–gene links: GeneHancer (Fishilevich et al., Database 2017).

Do biologically motivated inputs help?

The five versions compared on the next slides: same model, different inputs

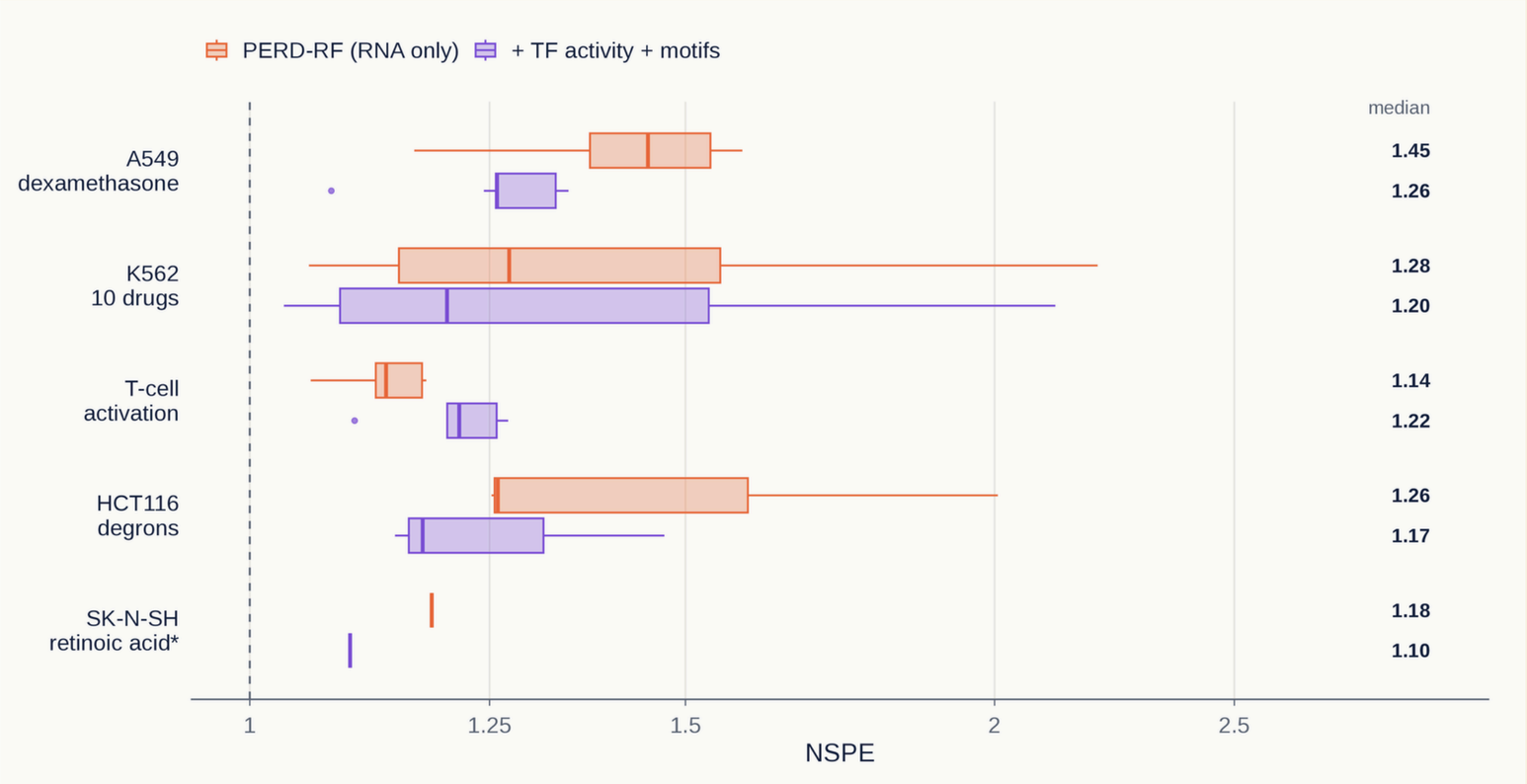
Input (per enhancer)	What it is	PERD-RF (RNA only)	+ shuffled activity (control)	+ TF activity	+ motifs	+ TF activity + motifs
mRNA of target genes	Genes the enhancer regulates (GeneHancer)	✓	✓	✓	✓	✓
mRNA of ChIP-bound TFs	TFs with a ChIP-seq peak in the enhancer	✓	✓	✓	✓	✓
Activity of ChIP-bound TFs	Same TFs, scored by their real target genes	–	–	✓	–	✓
Activity of motif families	TF families whose motif is common in the enhancer's DNA	–	–	–	✓	✓
Shuffled activity of ChIP-bound TFs	Same TFs, scored by random genes (control)	–	✓	–	–	–

Biology inputs help to improve the model



- **Better inputs lower the error on unseen tissues**
- Median NSPE 0.276 → 0.245 with TF activity and motifs together.
- Motifs give the clearest gain; TF activity alone is only slightly better than its shuffled control by NSPE.

The drug-induced change is still not predicted



- The new inputs lower the change error a little in most datasets but it stays above 1 everywhere.
- In T cells TF activity makes it worse (1.14 → 1.22).

Does training on real perturbations help?

- New model (pair model): one random forest that learns the change directly from real before/after pairs values (both predictors and target prediction)
- 16 perturbations: A549 dexamethasone, 10 K562 drugs, T-cell activation, 3 HCT116 degrons, SK-N-SH retinoic acid
- Inputs: all the additional inputs that were described previously + starting enhancer openness values
- Leave one perturbation out: train on 15, predict the 16th, repeat for all 16
- Control: starting enhancer openness values

Training on real perturbations still does not help



Median

- Over all 16 perturbations only the pair model (0.94) and the starting-openness control (0.92) that performed below 1 in NSPE.

Key Takeaways

RNA predicts where enhancers are open but predicting how drugs change them needs more

Validation	PERD predicts where enhancers are open, even in tissues it never saw, but not how much a drug changes them (NSPE > 1 in all five datasets).
Improvement	(1) TF activity + motifs as extra inputs; (2) a new model trained on real before/after pairs.
Result	Better inputs lower the error on unseen tissues (0.276 → 0.245), but barely help with drug change. The pair model is the first to beat "no change" (0.94), but starting openness alone does as well (0.92).
Limitation	Few, noisy perturbation datasets: 16 perturbations, mostly 2 replicates per side. Code reproduction is no 100% similar with the published paper.
Next	More replicated perturbation data; try different models.



Thank You!